

Problem Sheet 1

Two questions, 20 marks each — Total: 40 marks

Instructions. Use an AI assistant for the initial response to each question. Enter the supplied prompt exactly, in a new conversation for each question, and retain the first complete response. Do not regenerate responses until you obtain a mistake. For each question, record the assistant's name, the exact model label shown in its interface, and the date and time (including time zone). Also record whether web search or a special reasoning mode was enabled, and disclose any active custom instructions or personalisation known to you. If no model label is available, write "model not disclosed"; do not infer the model from its answer or ask it to identify itself.

Attach the exact prompt and complete, unedited response as a transcript or legible screenshots, together with a screenshot showing the assistant and model label where available. These records must be sufficient for another person to attempt the same request. Report what actually happened; do not invent, edit, or substitute an AI response.

Write your evaluations, source checks, and revisions yourself, without AI assistance. Use Chapters 4 and 5 of the lecture notes and sources that you personally open and read. Identify the relevant lecture-note sections; for outside sources, provide the organisation or author, page title, URL, and date accessed. An AI-generated citation is not verification.

Write your answers on A4 sheets, with your name and section, and label each question and subpart. Attach the AI records. No paid subscription is required; if you cannot access an AI tool, contact the instructor for an equivalent supplied response. Use only the fictional scenarios below; do not upload personal information or private research data.

How your work will be assessed. Marks reward accurate use of course concepts, specific evidence, and justified decisions. You do not have to find a mistake: credit is available for explaining why a recommendation is sound, what you checked, and where its limits lie. Critique the content, not merely the writing style or the fact that AI produced it.

1. (20 marks) Enter this prompt:

Four biology students are collecting plant observations and photographs for a group field project. They use personal phones, have limited mobile data, and sometimes have no internet connection. They also share access to a campus computer. They need to preserve the original observations, edit a shared data table, produce a report, and keep working if one member's account becomes unavailable. Recommend a practical digital workflow in no more than 250 words. Explain how the group should collect, organise, share, protect, and recover its files and how another member could check one reported result.

- (a) Submit the prompt, first complete response, and tool details required above. (2 marks)
- (b) Evaluate four specific recommendations or omissions: one concerning offline work, one concerning synchronisation and backup, one concerning sharing and account continuity, and one concerning preservation of raw data and reproducibility. For each, quote a short relevant excerpt or identify the omission, explain your judgment using Chapter 4, and state a practical consequence for this group. (8 marks)
- (c) Check one factual claim about a recommended application's or service's capabilities against its official documentation. State the claim, identify the documentation, report what it establishes, and explain any relevant condition or limitation. If no product is named, choose a tool that could implement one recommendation and verify the required capability. (4 marks)
- (d) Write an improved workflow in your own words. Explain two decisions to change or retain parts of the AI response, and describe a practical test of file recovery or reproducibility. (6 marks)

2. (20 marks) Enter this prompt:

Consider this fictional online post: “A laboratory study proves that a herbal drink prevents dengue.” The video has many views, a presenter wearing a laboratory coat, synthetic narration, and a link to buy the drink. It gives no study title, author, or link. After watching it, a biology student sees similar content in search results, recommended videos, and sponsored posts. In no more than 250 words, explain why the content may keep appearing, evaluate whether the information supplied supports the claim, and recommend what the student should check before sharing it. Do not invent the missing study or treat this fictional post as an identifiable real publication.

- (a) Submit the prompt, first complete response, and tool details required above. **(2 marks)**
- (b) Evaluate the response on four points: how it distinguishes search, recommendation, and advertising; whether it separates popularity and visual appearance from evidence; whether it distinguishes a laboratory finding from prevention in humans; and whether its proposed source checks are adequate. For each point, identify what the response says or omits and explain your judgment using Chapter 5. **(8 marks)**
- (c) Independently check one factual statement in the AI response using an appropriate source, such as an official platform explanation or public-health resource. State the exact claim, explain why your source is appropriate, identify the relevant information, and conclude whether it supports, contradicts, or leaves the claim unresolved. Then explain why this check cannot identify or validate the unspecified study in the fictional post. **(6 marks)**
- (d) Write your own short response to a classmate who wants to share the post. State your recommendation, give an evidence-based reason, identify what remains unknown, and explain what evidence would justify reconsidering your judgment. Do not give treatment advice. **(4 marks)**